ElevateSafe

Elevator Monitoring System

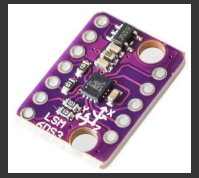
IoT: Algorithms and Services Group Project

Riccardo Passacantando - 1814248
Lorenzo Pellegrino - 2007948
Matteo Ciula - 2065405

Sapienza University of Rome

VERY Short Recap

- Low budget
- Minimal install required



LSM6DS3
6-axis IMU
+
AH3503
Hall sensor

Objectives

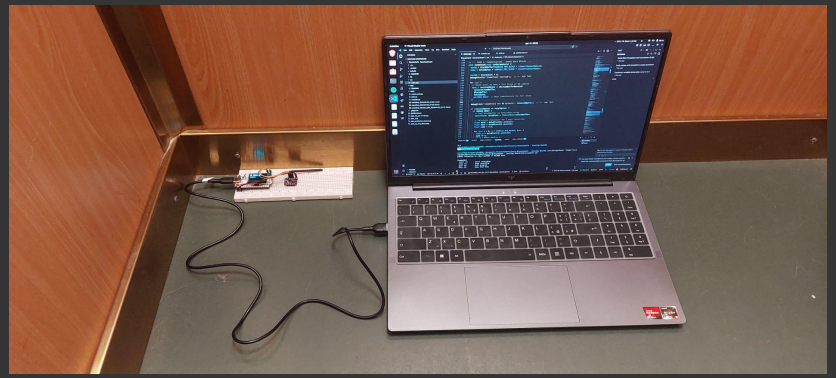
- Monitor behaviour of elevators to:
  - Warn if an unusual behaviour is detected
  - Optimize light to decrease energy consumption

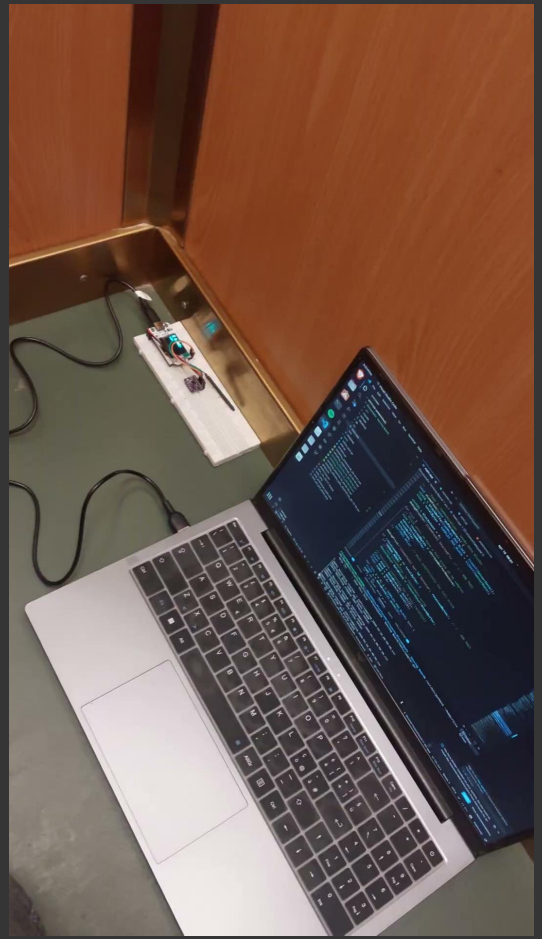
Implementation

- Training with raw data
- Inference on edge device
- LoRa communication to central node

Data collection

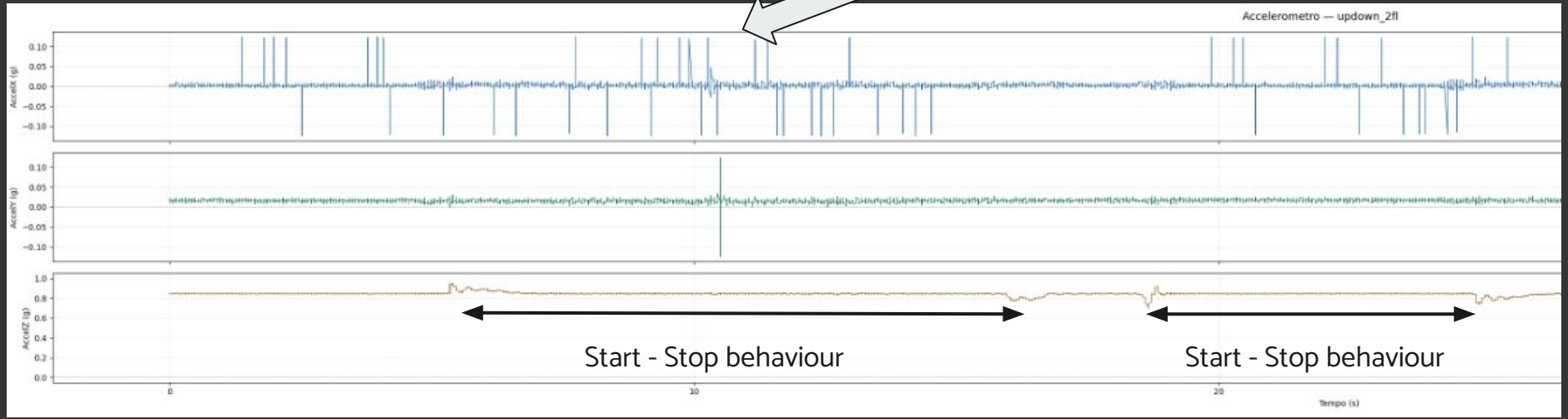
Raw data collection → Training for behaviour detection → Inference on edge → Alert sending



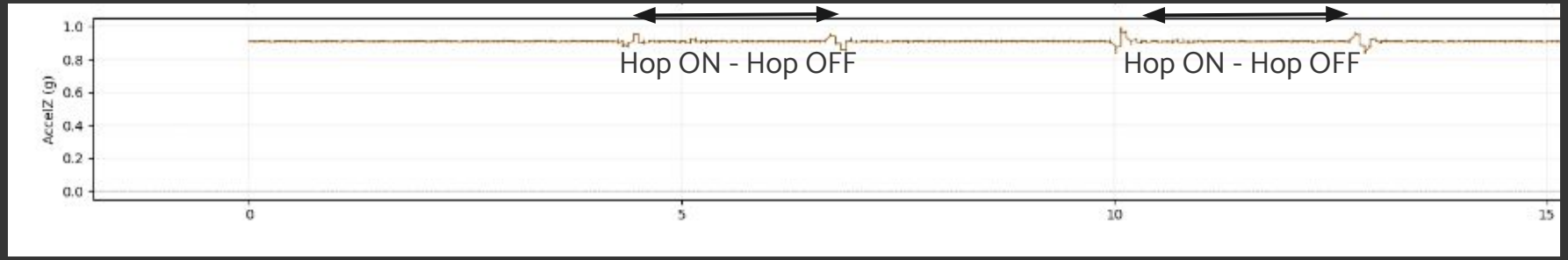


Raw data collection

Spikes on X values for vibrations



Light optimization



Detecting when a person hops on/off the elevator can lead us to analyze and recognize how to optimize light consumption

Energy analysis

COMPONENTS	WORST CASE	OPTIMIZED CASE
HELTEC V3	Always active: ~100 mA	No Radio + Active: **~ 50 mA** With Deep Sleep: **~ 12 mA**
LoRa SX1262	Always transmitting: ~120 mA	10 transm/h: **~ 0.02 mA**
IMU sensor	Always reading: ~4 mA	Wake on Motion: **~ 0.4 mA**
HALL sensor	Always reading: ~6 mA	Reading only on trigger (6 mA for 0.04s/h): **negligible**
1000mAh battery	**~230 mA** **~ 4h of usage**	**~ 13 mA** **~ 80h of usage**

Optimization

- **Duty Cycle Lora**
  burst transmission (50ms every 2s at 120mA) + standby (0.6µA)
- **Deep Sleep**
  Sleep when standing still at floor for more than 3 seconds
- **Power Management**
  Radio deactivation + IMU sensor interrupt

LoRa parameters

Spreading Factor	SF7
BandWidth	125 kHz
Coding Rate	4 / 6 (2 control bit)
Range	< 1 km

Payload with 25 samples of accelerometer data	150 byte
ToF	~258 ms

Sampling	1000 Hz
Gathering of 25 samples	~25ms

TOO FAST!

Sampling	50 Hz
Gathering of 25 samples	~500ms

Ok, but, we would have 10+ device simultaneously sending

SATURATION

LoRa parameters

Spreading Factor	SF7
BandWidth	125 kHz
Coding Rate	4 / 8 (4 control bit)
Range	< 1 km

Can increase CR for reliability

Sampling	1000 Hz
Gathering of 25 samples	~25ms

STILL FAST!

But, we will not have 10+ device simultaneously sending everytime

Payload with alert data	2 byte
ToF	~45 ms

Using Edge Computation to only send alerts